

IRIS dataset



Iris Versicolor



Iris Setosa



Iris Virginica

Implementation and Classification of Algorithms using Iris Dataset

We used the Iris dataset, which contains 6 columns:

Id, Sepal Length, Sepal Width, Petal Length, Petal Width, and Species, with a total of 150 rows.

The dataset is well structured and ready to work directly with machine learning algorithms. In this dataset, ['Sepal Length', 'Sepal Width', 'Petal Length', 'Petal Width'] are the input features, also called independent variables, which help in classifying the type of iris flower.

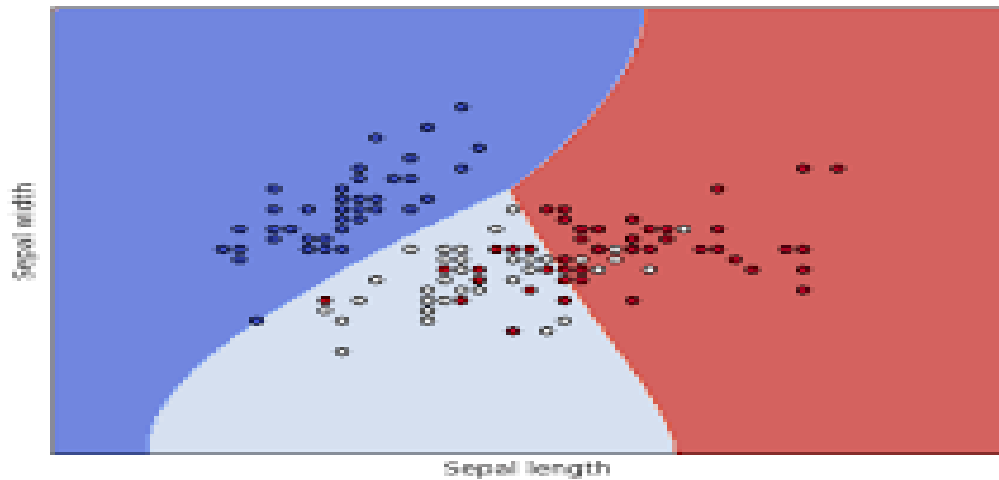
The output column is ['Species'], which contains 3 categories:

- Iris-setosa
- Iris-versicolor
- Iris-virginica

We performed feature scaling using Standard Scaler from sklearn. This method normalizes the dataset by converting all feature values to a common scale with mean 0 and standard deviation 1.

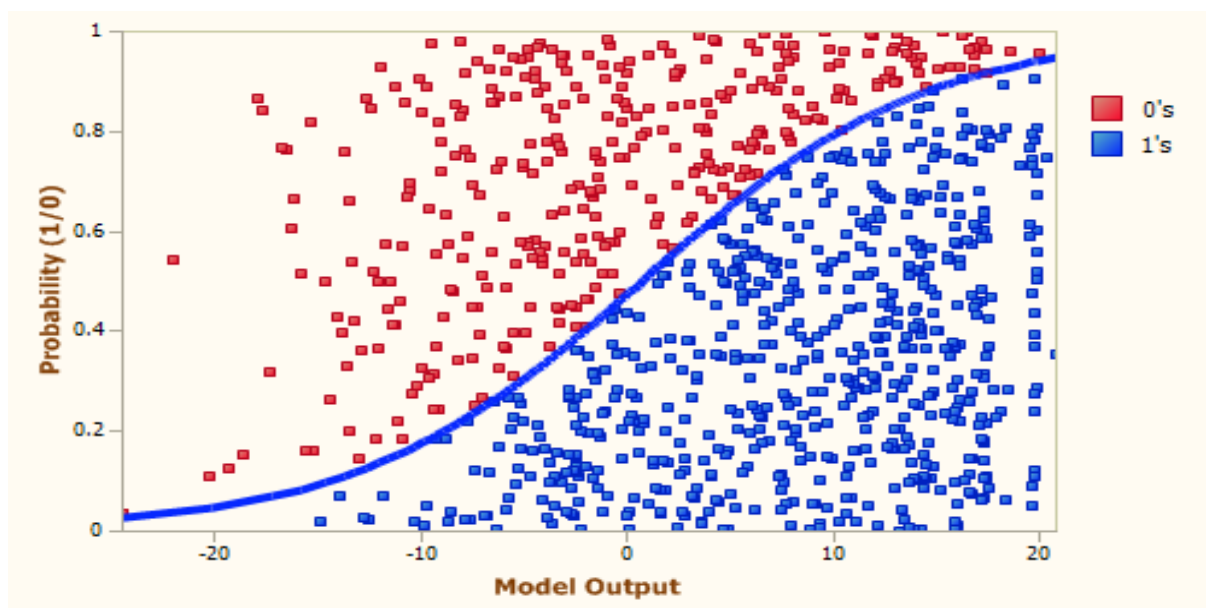
This helps in improving model performance and prevents bias caused by features having different value ranges. It is especially useful for algorithms like Support Vector Machine (SVM) and Logistic Regression.

Support Vector Machine (SVM):



Support Vector Machine (SVM) is a supervised machine learning algorithm mainly used for classification problems, and sometimes for regression. SVM performs very well when the dataset is clean and clearly separable. It is also highly effective in high-dimensional datasets and often provides high accuracy. We applied feature scaling using Standard Scaler before using SVM because SVM is sensitive to feature values with different ranges. Scaling improves the model performance and helps in finding the best decision boundary more accurately. Thus, Support Vector Machine is a powerful and accurate classification algorithm widely used in machine learning tasks like flower classification, spam detection, face recognition, and medical diagnosis.

Logistic Regression:



Logistic Regression is a supervised machine learning algorithm mainly used for classification problems. It is used when the output variable is categorical, such as Yes/No, True/False, or multiple classes like flower species classification. Unlike Linear Regression, Logistic Regression predicts the probability of a class belonging to a particular category using a sigmoid function. Based on this probability, the model assigns the final class label. We applied Standard Scaler before Logistic Regression because feature scaling improves the performance of the model and helps all features contribute equally during training.

Unlike Linear Regression, Logistic Regression predicts the probability of a class belonging to a particular category using a sigmoid (S-shaped) function. Based on this probability, the model assigns the final class label.

Gaussian Distribution:

Gaussian Distribution, also called Normal Distribution, is a continuous probability distribution that shows how data values are distributed around the mean (average).

It forms a bell-shaped curve where most of the values are concentrated near the center (mean), and fewer values are found as we move away from the center. In machine learning, Gaussian Distribution is commonly used in algorithms like Gaussian Naive Bayes, where it is assumed that the input features follow a normal distribution.

For example, in the Iris dataset, features like Sepal Length, Sepal Width, Petal Length, and Petal Width are assumed to follow a Gaussian distribution for each flower species.

Decision Tree Classifier:

Decision Tree Classifier is a supervised machine learning algorithm used for classification problems. It works like a flowchart, where data is divided into branches based on conditions and rules. The model starts with a root node and splits the dataset into smaller groups using important features. Each internal node represents a condition, each branch represents the outcome of that condition, and each leaf node represents the final class label. Decision Tree is easy to understand and visualize. It does not require feature scaling like SVM or Logistic Regression and works well with both numerical and categorical data.

However, it may sometimes overfit the training data if the tree becomes too deep. To avoid this, techniques like pruning and limiting tree depth are used. Decision Tree Classifier is widely used in applications like medical diagnosis, customer segmentation, fraud detection, and student performance prediction.

Random Forest Classifier:

Random Forest Classifier is a supervised machine learning algorithm used for classification problems. It is an advanced version of the Decision Tree algorithm. Instead of using only one decision tree, Random Forest builds multiple decision trees and combines their results to make a final prediction. This improves accuracy and reduces the problem of overfitting.

Each tree is trained on different random samples of the dataset, and the final output is decided using majority voting. The class that gets the highest number of votes becomes the final prediction.

Unlike SVM and Logistic Regression, Random Forest does not require feature scaling. It is widely used in applications like disease prediction, fraud detection, recommendation systems, and customer behavior analysis. Thus, Random Forest Classifier is a powerful and reliable algorithm for classification tasks with high accuracy and better generalization.

Conclusion:

In this, we applied multiple machine learning classification algorithms on the Iris dataset, including Support Vector Machine (SVM), Logistic Regression, Gaussian Naive Bayes, Decision Tree Classifier, and Random Forest Classifier.

The dataset was well-structured and contained important flower features such as Sepal Length, Sepal Width, Petal Length, and Petal Width, which were used as input variables to predict the flower species. Feature scaling was performed using Standard Scaler to normalize the data and improve the performance of algorithms like SVM and Logistic Regression. Each algorithm was trained and tested successfully, and accuracy scores were calculated to compare their performance.